

Stochastic Variational Inference (SVI) Assignment — Problems

ELG 5218 — Uncertainty Evaluation in Engineering Measurements and Machine Learning

University of Ottawa

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Context (given). We observe data $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ generated by

$$y_i = w x_i + b + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, s).$$

We fit the Bayesian linear regression model

$$\begin{aligned} w &\sim \mathcal{N}(0, 10^2), & b &\sim \mathcal{N}(0, 10^2), & s &\sim \text{HalfCauchy}(10), \\ y_i &| w, b, s, x_i \sim \mathcal{N}(w x_i + b, s). \end{aligned}$$

We approximate the posterior $p(\theta | \mathcal{D})$ over $\theta = (w, b, s)$ using NumPyro SVI. Unless stated otherwise, use $N = 60$ data points generated with $w_{\text{true}} = 0.5$, $b_{\text{true}} = -0.7$, $s_{\text{true}} = 1.0$, $x_i \sim \mathcal{N}(0, 5^2)$ (sorted), and $y_i = w_{\text{true}} x_i + b_{\text{true}} + \varepsilon_i$.

Conceptual Understanding (SVI & Variational Inference)

1.1 VI vs SVI (scaling).

- (a) In 4–6 sentences, explain the difference between (i) full-batch VI and (ii) SVI.
- (b) Let $N = 10^7$ and suppose the log-likelihood decomposes as $\log p(\mathcal{D} | \theta) = \sum_{i=1}^N \log p(y_i | x_i, \theta)$. Explain why full-batch VI requires $\Theta(N)$ work per gradient step.
- (c) In SVI, we use a minibatch $\mathcal{B}$ of size $M \ll N$. Write the unbiased estimator of the log-likelihood sum using $\mathcal{B}$.
- (d) In one sentence: what statistical/optimization tradeoff comes with minibatching?

1.2 ELBO and KL equivalence (proof).

- (a) Define the ELBO for a variational family $q_\nu(\theta)$.
- (b) Prove: $\log p(\mathcal{D}) = \mathcal{L}(\nu) + \text{KL}(q_\nu(\theta) \| p(\theta | \mathcal{D}))$.
- (c) Explain in 1–2 sentences why maximizing ELBO minimizes the KL divergence.

Results Interpretation & Diagnostics

2.1 Loss/ELBO curve: concrete patterns. You train SVI for 2000 epochs with learning rate 10^{-2} .

- (a) Sketch (describe) a healthy loss curve (negative ELBO) and explain why it indicates convergence.
- (b) Describe a curve indicating the learning rate is too high.

- (c) With a validation split, describe a pattern in training vs validation metrics that indicates overfitting.

2.2 Minibatching effects. Switch to minibatches $M = 32$.

- (a) Write the minibatch estimator of the expected log-likelihood term.
- (b) Predict how the loss curve changes.
- (c) Give two stabilization methods.

2.3 Parameter trace diagnostics. You record $\mu_w(t)$, $\sigma_w(t)$, $\mu_b(t)$, $\sigma_b(t)$ over epochs.

- (a) Give one trace pattern suggesting poor initialization.
- (b) Give one pattern suggesting convergence to a suboptimal solution.
- (c) Provide two remedies.

2.4 Posterior plots: underfitting, overconfidence, mean-field failure.

- (a) Underfitting: what happens to $q(s)$ and the predictive band?
- (b) Overconfident posterior: what do marginals look like relative to w_{true} and b_{true} ?
- (c) Mean-field failure: what shape in joint (w, b) indicates correlation the guide cannot represent?

2.5 Quantitative evaluation with known ground truth.

- (a) Define bias estimate for w using posterior samples.
- (b) Define uncertainty for w .
- (c) Propose a held-out predictive metric.

2.6 Predictive bands interpretation.

- (a) Too narrow bands but many points outside: what does that suggest?
- (b) Too wide bands everywhere: what does that suggest?
- (c) Why should uncertainty increase away from training inputs?

Mathematical Derivations

3.1 ELBO for Bayesian linear regression.

- (a) Expand $\log p(\mathcal{D}, \theta)$ into log priors and log likelihood.
- (b) Write $\mathcal{L}(q)$ as expected log prior + expected log likelihood + entropy.

3.2 Log-normal noise guide and KL in log space. Let $z \sim \mathcal{N}(\mu, \sigma^2)$ and $s = \exp(z)$.

- (a) Derive $q(s)$.
- (b) For $s \sim \text{HalfCauchy}(\gamma)$, derive $p(z)$ up to proportionality.
- (c) Write $\text{KL}(q(z) \| p(z))$ as an expectation under $q(z)$.

3.3 Gradient of ELBO w.r.t. μ_w . Use $w = \mu_w + \sigma_w \epsilon$, $\epsilon \sim \mathcal{N}(0, 1)$. Derive an expression for $\nabla_{\mu_w} \mathcal{L}$.

3.4 AutoDelta and MAP. Show that a point-mass guide corresponds to MAP optimization and explain why it is degenerate.

Comparative Analysis

4.1 SVI vs HMC/NUTS (structured comparison). Write a comparison that addresses each bullet below with 2–3 sentences:

- (a) *Computational cost*: per-iteration cost and scaling with dataset size.
- (b) *Accuracy*: approximation error vs asymptotic exactness.
- (c) *Posterior correlations*: ability to represent and recover dependencies.
- (d) *Robustness*: sensitivity to tuning and model geometry.

4.2 Why can AutoMultivariateNormal be better but slower? Give two concrete technical reasons and connect each to a measurable effect.

- (a) Parameter count / compute.
- (b) Optimization difficulty.